

SELECTED WORK / SEPTEMBER 2026

AI & Data Portfolio

Ruddi Rodriguez Garcia

AI agents and reliable integrations

Predictive analytics and reusable data workflows

Scientific computing and clear, practical outputs

Selected existing presentations. Historical results and illustrative examples are labelled.

Source example: adaptive interview agents with fictional data and an offline walkthrough.

github.com/RuddiRodriguez/freelance-ai-portfolio

Git history verified

Amazon Bedrock Integration for a Shared LLM Framework

Historical Converse architecture adapted to one provider-neutral application contract

APPLICATION INTERFACE

1 contract

Provider-neutral framework

SUPPORTED INPUTS

Messages + tools

Shared request structure

PROVIDER ROUTES

2 paths

Converse + compatible Responses

UNIFIED OUTPUTS

Text + JSON

Tools, usage and reasoning

HYBRID PROVIDER ARCHITECTURE

Application contract

Conversation + system message

User and assistant messages

Functions and tool choice

Strict JSON schema

Primary + fallback model

LLM Handler

One prediction interface

Provider selection

Fallback without mutation

Common request options

Error propagation

Native Bedrock models

Messages, tools and schemas translated to Amazon Bedrock Converse format.

Bedrock-hosted OpenAI models

Region-aware Bedrock endpoint using compatible OpenAI Responses contract.

Unified result

Assistant message

Parsed tool arguments

Structured output

Finish reason

Token usage + reasoning

Key outcome: downstream applications kept the same conversation and result objects while provider-specific formats, credentials and endpoints stayed inside the Bedrock client.

CONVERSE ADAPTATIONS

- Message conversion**
 System and conversation roles mapped to Bedrock content blocks.
- Tool translation**
 Framework functions converted to toolSpec, inputSchema and toolChoice.
- Structured output**
 Strict JSON schemas normalized for Bedrock output configuration.
- Response normalization**
 Text, toolUse, reasoningContent and token usage returned in common objects.
- Compatibility handling**
 System-message and tool limitations handled per model capability.
- Failure recovery**
 Provider errors classified and fallback preserved through LLM Handler.

HISTORICAL ARCHITECTURE · CONVERSE PATH LATER REMOVED

Portfolio summary uses repository history and excludes proprietary prompts, credentials and client data

Customer response ranking and classification framework

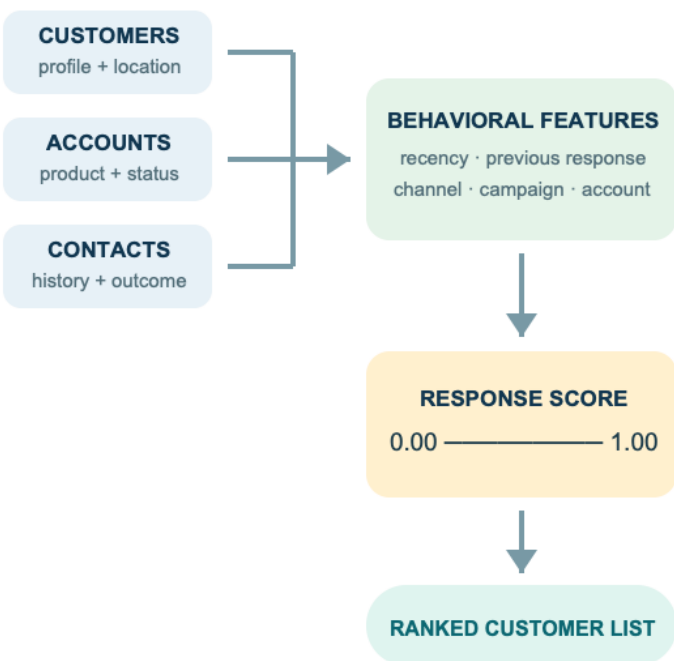
Machine learning case study

From raw customer history to ranked, measurable campaign decisions

data preparation · model selection · validation · production scoring

1 · Business problem

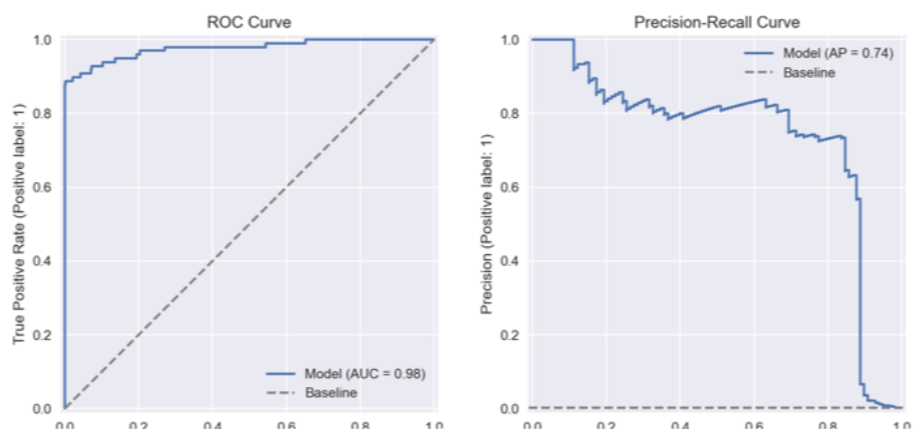
Contact capacity is limited. Every customer receives a response probability, then campaigns focus on the highest potential records instead of treating all customers equally.



2 · Reusable model workflow



Stored validation example from reusable classifier



0.98

ROC AUC

0.74

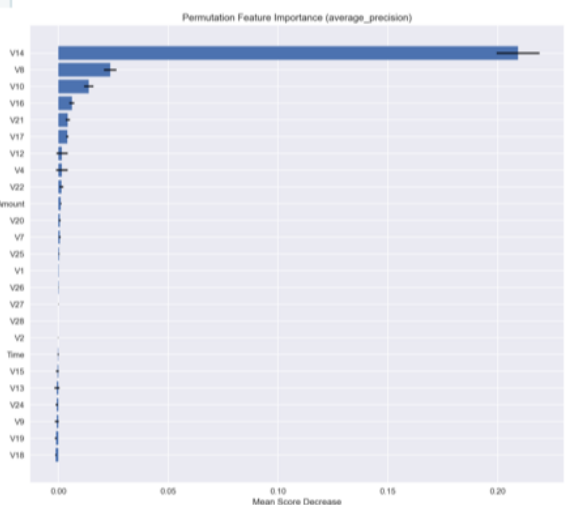
AVERAGE PRECISION

0.887

KS SEPARATION

Validation example uses anonymized public-format data. Original customer records are not shown.

3 · Explain and deliver



Operational outputs

- ranked records at chosen cutoff
- lift, gain and threshold analysis
- saved production scoring pipeline
- interactive validation dashboard

Delivered system

SQL data preparation → reusable preprocessing → model comparison → tuning → validation → probability ranking → campaign selection

PRIVATE REPOSITORY

portfolio visual excludes source customer data and internal paths

Broken LLM workflow → reliable structured output

48 LOCAL TESTS PASSED

API repair, strict JSON validation, safe fallback routing, and focused tests

1 Diagnose

Unstable response

```
Here is the result:
```json
{ priority: high, ... }
```
```

PARSER ERROR

- mixed prose and JSON
- missing required fields
- stale request options

2 Repair

- ✓ Match the current API contract
- ✓ Define a strict JSON schema
- ✓ Keep the same schema on fallback
- ✓ Separate recoverable failures

```
+schema = {"type": "object",
+  "required": [
+    "priority", "summary"
+  ],
+  "additionalProperties": false
```

3 Verify

Validated result

```
{
  "priority": "high",
  "summary": "Payment API failed",
  "needs_human": true
}
```

SCHEMA VALID

- ✓ Predictable application data
- ✓ Tested failure and fallback paths

Sanitized demonstration based on a working Python implementation — no client data shown

Stochastic Life Science Simulation Framework

Event-driven modeling of complex biological systems using Gillespie sampling

Python · Computational Biology · Mathematical Modeling
 Reusable methodology for stochastic biological processes

Reproducible simulation design

SIMULATION METHOD

Direct Gillespie

Continuous-time stochastic events

SYSTEM BEHAVIOR

Fixed + dynamic

Static and evolving structures

REPRODUCIBILITY

Seeded ensembles

Repeatable random experiments

COMPUTATION

Parallel runs

Independent simulations across CPUs

SCIENTIFIC MODELING PIPELINE

01

Biological mechanism

Define molecular interactions, physical constraints and experimentally observed behaviors.

02

Model parameters

Convert rates, forces, diffusion and geometry into named configurable inputs.

03

Event system

Represent attachment, detachment, movement, growth and state changes as explicit events.

04

Stochastic engine

Sample next-event time and identity from current state-dependent propensities.

05

Ensemble analysis

Aggregate independent traces into distributions, uncertainty and condition comparisons.

State-dependent rates

Event probabilities change as biological state evolves.

Random waiting time

Continuous-time delay sampled from total propensity.

Weighted event choice

Next mechanism selected according to relative rates.

Iterative state update

Every event becomes input for next simulation step.

Molecular kinetics

Binding, unbinding and reaction-rate processes.

Cellular mechanics

Force-dependent behavior and physical constraints.

Dynamic structures

Growth, shrinkage, transitions and moving boundaries.

Condition comparison

Control versus perturbed biological scenarios.

GILLESPIE SIMULATION CYCLE

Calculate rates

Read current biological state.

Sample time

Draw stochastic waiting interval.

Select event

Use cumulative propensities.

Update state

Apply mechanism and constraints.

METHODOLOGICAL STRENGTHS



Mechanism-first design

Named events make biological assumptions visible and auditable.



Analytical connection

Model equations and stochastic simulation share one parameter source.



Reproducible randomness

Master seeds create unique, repeatable traces across workers.



Parallel ensembles

Independent traces scale across CPU processes without changing event order.



Scientific safeguards

Boundary checks preserve physical and biological constraints.



Transparent assumptions

Unavailable measurements and substitute parameters remain explicitly documented.



Analysis-ready outputs

Traces, categories, means and uncertainty summaries support reporting.

GENERALIZED LIFE-SCIENCE METHODOLOGY

Designed for reproducible research · No proprietary experimental data shown · Results depend on supplied assumptions

Hybrid AI + Deterministic Stock Paper-Trading Agent

Alpaca Paper through SnapTrade • Continuous Python agent • Research-to-execution audit trail

Ruddi Rodriguez Garcia
Designed, built, tested, and operated
Python • ML forecasting • LLM agents • Broker execution

Separation of responsibilities: LLM discovers and interprets. Deterministic engine validates evidence, selects models, sizes capital, and controls orders.

Paper execution only

1. Broker snapshot

Fresh Alpaca Paper cash, equity, holdings, profit/loss, open orders, permissions, and reserved capital. One normalized source of truth for whole cycle.

2. Existing holdings first

Forecast, company-quality review, loss/concentration checks, event risk, and open-order review. HOLD, ADD, TRIM, EXIT, or REVIEW. Missing evidence forces HOLD.

3. Sector allocation

Portfolio exposure, cash, market context, diversification, and opportunity themes produce bounded sectors for candidate discovery.

4A. Deterministic selector

- Tradable stock universe and sector coverage
- Price history, liquidity, trend, volatility, and data quality
- Existing holdings and open-order exclusions
- Fails closed when universe coverage is incomplete

4B. LLM selector

- Current news, catalysts, fundamentals, and market regime
- Candidate rationale, main risks, and source notes
- Structured output; no broker access and no order authority
- Uses deterministic shortlist as grounded research context

5. Combined shortlist

Merges independent selector evidence, removes current positions and conflicting orders, ranks new candidates, and records provenance for every ticker.

Source trace

Portfolio-aware

No silent fallback

Bounded candidates

6. Forecast competition

Adaptive classical model shortlist by ticker and horizon. Walk-forward validation, transaction costs, slippage, directional accuracy, Sharpe, drawdown, and leakage checks select champion.

7. Finance & catalyst review

LLM finance expert examines business quality, balance sheet, red flags, catalysts, confidence, and source support. Interpretation cannot override failed model or data gates.

8. Portfolio & capital decision

Candidate impact, concentration, available cash, open-order reservations, expected return, execution confidence, buy zone, and per-order limits produce WAIT or approved limit plan.

9. Paper execution & feedback

Market-open check, fresh quote, broker preview, GTC single or ladder limit order, journal, duplicate/stale-order reconciliation, final broker snapshot, and model-performance feedback.

Illustrative decision trace

Candidate XYZ

→ Selectors pass

→ Forecast edge

→ Finance pass

→ Risk pass

→ Broker preview

→ Paper limit

Any failed gate: WAIT / HOLD / no order. Failure stays visible in report.

Operational cadence: premarket analysis → market-open submission → continuous open-order state checks → next-cycle learning.

Architecture based on working stock paper-agent code. Example ticker is fictional. No account data, credentials, positions, or proprietary parameters shown.

AI-Enabled Multi-Broker Trading and Market-Intelligence Platform

Ruddi Rodriguez Garcia
Python • Algorithmic Trading • AI/ML
IBKR • Alpaca • SnapTrade • Deribit

Research → validation → deterministic risk → broker execution → measured feedback



Core design principle: AI supports research and strategy review. Deterministic Python owns eligibility, sizing, risk, orders, and broker safety.

1 Market & Broker Data

Point-in-time market inputs

Prices, volume, volatility, curves, expiry, trading hours, corporate and macro events.

Broker truth

Accounts, equity, buying power, positions, open orders, exact contracts, quotes, and margin previews.

News & catalysts

Deterministic RSS collection, IBKR contract news, company events, supply-demand and geopolitical drivers.

IBKR News Yahoo RSS



2 Research & Strategy

AI research agents

News validation, catalyst analysis, fundamentals, market regime, portfolio context, and structured research reports.

Forecasting & candidate generation

Classical ML, time-series models, pure-LLM analysis, sector selection, and strategy candidates.

Strategy evaluation

Walk-forward validation, leakage checks, transaction costs, slippage stress, Sharpe, drawdown, and benchmark comparison.

OpenAI Bedrock MLflow



3 Deterministic Control

Portfolio and position sizing

Equity, available cash, margin, contract multiplier, exposure, concentration, and risk-budget controls.

Trade-governance gates

Market hours, liquidity, spread, expiry and roll windows, duplicate orders, stale data, and incomplete-evidence safe stops.

Paper/live separation

Explicit execution flags, account permissions, previews, immutable audit records, and fail-closed behavior.

Risk first Auditable Fail closed



4 Broker Execution

Native IBKR commodity futures

Exact contract discovery, limit and stop orders, what-if margin, expiry/roll controls, and broker-held GTC protection.

Multi-broker trading

Alpaca and SnapTrade stocks/options; Deribit crypto derivatives; normalized account and order state.

Order lifecycle

Preview, submit, monitor, reconcile, modify, cancel, fill handling, stale-order decisions, and duplicate prevention.

IBKR Alpaca SnapTrade Deribit

Continuous risk protection

60-second IBKR position checks plus broker-native protective stops that remain active when agent disconnects.

Daily research cycle

Market news, contract data, portfolio state, allocation review, and approved paper-order planning.

Reconciliation & feedback

Orders, fills, stops, positions, and strategy outcomes feed back into next evaluation cycle.

Monitoring & audit

JSON reports, dashboards, decision traces, broker responses, model cost, and execution blocks.

Selected commodity coverage: crude oil, natural gas, gold, silver, copper, corn, wheat, soybeans.

Architecture based on working private repositories and operating paper-trading agents. No account data or proprietary strategy parameters shown.

AI Stock Forecasting & Risk Analysis Agent

Structured forecast complete

CURRENT PRICE

€126.81

Historical input

AGENT DECISION

WAIT

Do not chase

PREFERRED ZONE

€126.00–
126.20

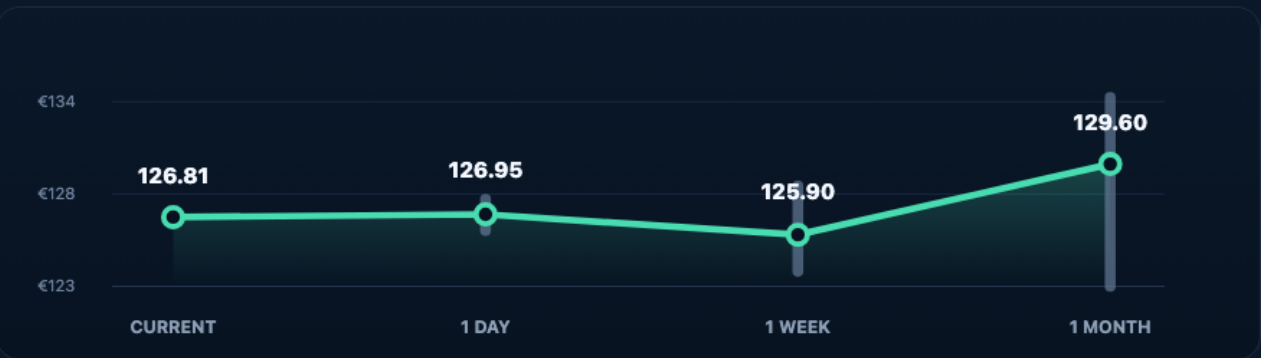
Stabilization required

ZONE PROBABILITY

80.1%

10-day touch estimate

MULTI-HORIZON PREDICTED PRICE



| HORIZON | PREDICTION | BEAR–BULL RANGE | CONFIDENCE | DIRECTION |
|---------|------------|-----------------|------------|-----------|
| 1 day | €126.95 | €126.10–127.80 | 54% | FLAT |
| 1 week | €125.90 | €123.90–128.70 | 56% | DOWN |
| 1 month | €129.60 | €123.00–134.20 | 58% | UP |

DECISION SUMMARY

WAIT

Buy only after stabilization near €126.00–126.20 or confirmed reclaim above €128.00. Current price offers weaker risk/reward.

| | |
|-----------------------|------------|
| Breakout trigger | €128.00 |
| Invalidation | €123.80 |
| Take-profit reference | €129.60 |
| Probability source | Historical |

CONFIDENCE LEVEL · MEDIUM

LOW MEDIUM HIGH

AGENT PIPELINE

| | |
|--------------------------------|---|
| 01 · Price history | ✓ |
| 02 · Market & macro context | ✓ |
| 03 · Technical and risk layers | ✓ |
| 04 · Structured LLM forecast | ✓ |

RESEARCH ONLY · ADVICE IS NOT AN ORDER

Broker submission disabled · OpenAI + local MLX compatible · Historical result, not current investment advice

LIVE DEMO / DUTCH → ENGLISH

```
$ python realtime_translator.py --src-lang nl --dest-lang en --show-source
```

[source] Goedemorgen, de vergadering begint om tien uur.

[en] Good morning, the meeting starts at ten o'clock.

Portfolio demonstration • synthetic voice input • commands from working prototype